
UNIT 15 CENTRAL BANKS AND THE SUPPLY OF MONEY*

Structure

- 15.0 Objectives
- 15.1 Introduction
- 15.2 Money Supply
 - 15.2.1 Measurement of Money
 - 15.2.2 Money Multiplier
 - 15.2.3 An Alternative Interpretation of Money Multiplier
- 15.3 Open Market Operations
- 15.4 Let Us Sum Up
- 15.6 Answer/ Hints to Check Your Progress Exercises

15.0 OBJECTIVES

After going through this Unit you should be in a position to

- discuss how the supply of money is measured;
- identify the components of money supply;
- explain how changes in money supply take place; and
- analyse the scope of open market operations.

15.1 INTRODUCTION

Many years ago, around 6000 BC, when money was not invented, there was the barter system. Under this system, people used to trade goods and services with one another in the absence of money. As money was not invented, people exchanged goods and services for other goods and services in return. The disadvantage of this system was that one could exchange items only when there was a 'coincidence of wants' (also called double coincidence of wants). For example, I have wheat with me and I want apple in exchange. I should look for a person who has apple and wants to exchange it for wheat. This sort of coincidence, as you could imagine, was rare and created difficulties in transaction of goods and services. When money got invented, exchange process became smooth enough.

Money is anything that is commonly accepted by people for the exchange of goods and services. Every country has its own system of currencies in the form of coins and paper notes. Money in its present form however is a recent phenomenon. In the olden days, there were other entities which were considered as money. There was a lot of experimentation in the search for an ideal commodity which could serve the purpose of money. In this experimentation

* Dr. Jagannath Mallick, Independent Researcher, Delhi

there were animals, valuable commodities and precious metals. Salt, seeds, cattle, tea, etc. were popular commodities to be used as money in the past. Commodity money provided intrinsic value, but it faced certain difficulties in terms of transportation, storage, and lack of longevity. This led to introduction of coins made of precious metals such as gold and silver as money. Finally, non-precious materials have been used to make money, as we see today (such as coins and paper currencies). These are known as fiat money – they have neither intrinsic value nor can they be redeemed for any precious commodity. A currency note can be exchanged only with currency notes of other denominations.

There are certain criteria that money should fulfil, viz., (i) it should be amenable to standardisation so that all units are uniform, (ii) it should be acceptable by all, (iii) it should be divisible so that it can be exchanged with currencies of other denominations, (iv) it is easy to carry, and (v) it has certain longevity. Commodity money did not fulfil most of these criteria while the present day paper money fulfils all the criteria.

Money has three important functions to play, viz., (i) unit of account, (ii) medium of exchange, and (iii) store of value. Price of a commodity, which represents its market value, is expressed in terms of money. Thus money acts as a unit of account or a measure of value. Further, money can be used any time, at present or in future, to purchase any commodity. Thus it can be stored, from the time it is received till it is spent on purchase of some goods and services. Thus, having money entitles a person to certain purchasing power. Since each and every commodity is expressed in terms of price, it is easier to exchange goods and services. In fact, every transaction, whether selling or buying, can be seen as an exchange between the commodity concerned and money. Thus, money eliminates the problem of coincidence of wants and the process of exchange of commodities becomes smooth.

15.2 MEASURES OF MONEY SUPPLY

Money is issued by the Central Bank of the country concerned. In India, for example, the Reserve Bank of India (RBI) issues currency notes of several denominations. Coins of several denominations are issued by the Government of India, but circulated by the RBI. The RBI is responsible for design, production and management of currencies in the country. By money supply we mean the total amount of currency and other liquid instruments circulating in the country.

You should note that there is no unique definition of ‘money supply’, either as a concept in economic theory or as measured in practice. Money is defined by the functions it performs. Thus when we look into its ‘store of value’ function, there could be several instruments of storing money – as cash in hand, as savings deposit in a bank, as fixed deposit in a bank or post office, etc. Remember that each of these methods have different levels of liquidity. Similarly, acceptability of money as a ‘medium of exchange’ is not limited to currency in circulation. You can make payments through several methods such as cash, cheque and electronic transactions – each one being different in terms of liquidity. Remember that cheque and electronic transactions are backed by fiat money.

Since these instruments perform the functions of money, it is not appropriate to restrict money supply to currencies in circulation only.

Total stock of money includes all the money held with the public, the RBI as well as the government. By money supply we mean the first component only (that is, money held with the public). Here, the word ‘public’ includes the households, firms, local authorities, and companies. Money held by the RBI and the government are not included in money supply. Thus money supply does not include money held as cash reserve ratio (CRR) with the RBI, statutory liquidity ratio (SLR) with the commercial banks, and money held by the government as these are out of circulation.

Let us look into the definition of various measures of money stock in India. Since July 1935, the RBI is the sole authority to compile and disseminate the statistics on monetary indicators in India. Based on the changes in the structure of the Indian economy various working groups were set up over the years to review and redefine the monetary aggregates. The First Working Group on Money Supply (FWG) was set up in (1961), the Second Working Group (SWG) was formed in (1977) and the “Working Group on Money Supply: Analytics and Methodology of Compilation” (WGMS) was formed in (1998). The Monetary aggregates are published on a regular basis in various publications of the RBI such as Annual Report, Handbook of Statistics, and the RBI Bulletin.

From policy perspectives, money can be defined as the set of liquid financial assets, whose variation in the stock could influence aggregate economic activity. The quantification and study of the money supply directs the policy makers and economists to design the policy of changing money supply according to the situations in an economy. Thus, a range of monetary and liquidity measures are compiled in India (see Table 15.1 for their definitions).

Table 15.1: Measures of Monetary and Liquidity Aggregates

M0	=	Reserve Money
	=	Currency in circulation + Bankers’ deposits with RBI + ‘Other deposits’ with the RBI
	=	Net RBI credit to the Government + RBI credit to the commercial sector + RBI’s claims on banks + RBI’s net foreign assets + Government’s currency liabilities to the public – RBI’s net non-monetary liabilities
M1	=	Currency with the public + Demand deposits with the banking system + ‘Other deposits’ with RBI.

M2	=	M1 + Savings deposits of post office savings banks
M3	=	M1+ Time deposits with the banking system
M4	=	M3 + All deposits with post office savings banks (excluding National Savings Certificates)
NM0	=	Monetary base
	=	currency in circulation + Bankers deposits with RBI + 'other deposits' with RBI
NM1	=	Currency with the public + Demand deposits with the banking system + 'Other deposits' with the RBI
NM2	=	NM1 + Short-term time deposits of residents (including and up to the contractual maturity of one year)
NM3	=	NM2 + Long-term time deposits of residents + Call/ Term funding from financial institutions
L1	=	NM3 + All deposits with the post office savings banks (excluding National Savings Certificates)
L2	=	L1 + Term deposits with term lending institutions and refinancing institutions (FIs) + Term borrowing by FIs + Certificates of deposit issued by FIs
L3	=	L2 +Public deposits of the non-banking financial companies

Notes: *Net Bank Credit to the Government* = Net RBI Credit to the Government (i.e., Net RBI Credit to the Centre + Net RBI Credit to State Governments) + other banks' credit to the Government

Bank Credit to the Commercial Sector = RBI credit to the commercial sector + other banks' credit to the commercial sector

Net foreign assets of the banking sector = RBI's net foreign assets + other banks' foreign assets

Net non-monetary liabilities of the banking sector = RBI's net non-monetary liabilities + Net non-monetary liabilities of other banks.

Source: RBI

You should note that M0 is known as 'monetary base', 'reserve money', 'central bank money', or 'high-powered money'. The monetary measure M1 includes M0, demand deposits with commercial banks, and other deposits with the RBI (such as deposits of other financial institutions apart from banks, internal amount pending adjustments, etc.). The measures M0 and M1 are sometimes called 'narrow money' as they indicate money supply that can be easily converted into cash. If we add post office saving bank deposits to M1, we obtain M2. Further, if we add time deposits (also called fixed deposits) to M1 we obtain M3. If we add total post office deposits to M3, we obtain M4.

While M0, M1 and M3 are widely used in India, M2 and M4 are rarely used. The RBI started publication of new set of monetary aggregate (NM0 to NM3) and liquidity aggregates (L1 to L3) following the suggestions of WGMS in 1998. The measures NM2 and NM3 are applicable to the residents of the country. Currency and deposits held by non-resident Indians are normally recorded in the balance of payments accounts as international capital flows. However, it may not be appropriate to exclude all categories of non-resident deposits from domestic monetary aggregates as non-resident rupee deposits are essentially integrated into the domestic monetary market. Hence, the non-resident repatriable foreign currency fixed deposits such as foreign currency non-resident (bank) or FCNR (B), Resurgent India Bonds (RIBs), and India Millennium Deposits (IMDs) are excluded from deposit liabilities and they are treated as external liabilities. Other features of the new monetary and liquidity aggregates are as follows:

1. There are the four measures of monetary aggregates (NM0, NM1, NM2, NM3) and three measures of liquidity aggregates (L1, L2, L3) in India.
 - a) NM0 is the monetary base.
 - b) The NM1 is the non-interest bearing monetary liabilities of the banking sector
 - c) NM2 represents the transactions balances of entities.
 - d) NM3 includes NM2, long-term time deposits of residents and the call funding that the banking system obtains from other financial institutions.
2. Bank credit affects directly the consumption and capital formation. Provision of adequate flow of credit to the productive sectors is one of the important objectives of the monetary policy of RBI. The bank credit to the commercial sector includes advances in the form of loans, cash credit,

overdrafts, bills purchased and discounted, and investments in approved securities other than government securities. Further, in recent years, commercial banks are investing in various securities of the commercial sector like commercial paper, shares and debentures.

3. Net foreign exchange assets (NFA) of the banking sector include net foreign exchange assets of RBI and foreign assets of other bank. The latter include banks holdings of foreign currency assets net of i) their holdings of non-resident repatriable foreign currency fixed deposits which is presently defined to include FCNR(B) deposits, and ii) overseas foreign currency borrowings.

Check Your Progress 1

- 1) State the various aggregate measures of money supply in India.

.....

.....

.....

.....

- 2) Explain the differences between NM0, NM1, NM2 and NM3.

.....

.....

.....

.....

- 3) What are the liquidity aggregates in India?

.....

.....

.....

.....

15.3 MONEY MULTIPLIER

The commercial banks in an economy operate on the basis of fractional reserve system. Suppose you deposit Rs. 1000 in a bank. The bank has to deposit a fraction of it with the central bank (In India, for example, the cash reserve ratio is 4.5 per cent in 2022; thus the bank has to deposit Rs. 45 with the reserve bank of India). Further, the bank has to retain certain percentage in the form of cash, gold and securities. This percentage is called statutory liquidity ratio (SLR). The SLR as of 2022 in India is 18 per cent, so that the bank has to maintain a SLR of Rs. 180. The bank knows that all the customers will not withdraw their savings at the same time. Therefore, the bank extends credit to its customers out of the remaining amount, i.e., 775. When individuals receive Rs. 775 as loans, and spend it, there is an increase in income of some individuals to the extent of Rs.

775. These individuals deposit part of this Rs. 775 in banks. The bank, on receiving the deposit, is in a position to extend further loans, and the process goes on.

Thus, the monetary base expands to an effective money supply through bank lending. The excess of the deposits of the general public over bank reserves is lent out in the form of bank loans, or government or corporate bonds. The non-bank public then deposits a fraction of these loans on checking accounts. Next, the banks lend out a fraction of these and so on. This process is called the money multiplier process. The ratio of the money stock measured as M1 to the monetary base, M0, is defined as the money multiplier, which is also equal to the inverse of the 'reserve ratio' or 'cash reserve ratio (CRR)'. Let us assume that the required reserve ratio is 25 per cent of total deposits (= demand deposits + time deposits, which is the liabilities of the banks) by public with the banks. So, the money multiplier is $\frac{1}{0.25} = 4$.

15.3.1 Mathematical Derivation of Money Multiplier

For mathematical representation, let us assume

C = currency held by the general public,

D = demand deposits held by the general public,

$C/D = cd$, the desired currency deposit ratio (It implies, $C = cd D$)

R = bank reserves = currency held by the commercial banks plus their demand deposits in the central bank,

$R/D = rd$; the required reserve ratio (It implies, $R = rd D$)

It is important to mention that cd is chosen by the non-bank public, whereas rd is determined by the central bank in the form of CRR and excess reserves in India. The reserve ratio is meant to maintain liquidity in the economy. In addition, banks may hold excess reserves according to their requirements of liquidity and lending risks.

$$\text{Now, } M1 = C + D = cd D + D = (cd + 1)D \quad \dots (15.1)$$

$$\text{Monetary base } = M0 = C + R = cd D + rd D = (cd + rd)D$$

$$\text{Therefore, } D = \frac{M0}{cd + rd} \quad \dots (15.2)$$

Substituting the value of D from (15.2) in (15.1), we obtain

$$M1 = \frac{cd + 1}{cd + rd} M0 = m M0$$

$$\text{where } m = \frac{cd + 1}{cd + rd} \quad \dots (15.3)$$

The term m in equation (15.3) is the 'money multiplier'. It indicates the multiple by which money supply will increase, given the monetary base of the country. Suppose, for example, $cd = 0.4$ and $rd = 0.1$. In this case the money multiplier would be $m = 2.8$.

There are four implications of equation (15.3).

- a) Money multiplier will be higher if we consider a broader measure of money supply (say, M3 in place of M1).
- b) It is important to mention that m depends upon the value of currency-deposit ratio (cd) and reserve ratio (rd). Any change in the value of these two parameters will change the value of m . The parameters cd and rd depend on various factors such as expected returns, degree of liquidity, and risk on alternative assets from the customers and bank's perspectives.
- c) We can say that $\Delta M1 = m\Delta M0$. If we know the value of m , we can estimate by what extent M0 should be increased for a desired level of M1.
- d) For given values of M0 and cd , the money supply will decrease with higher reserve-deposit ratio (rd) and vice versa. Now you can relate how changes in CRR and SLR influence money supply. An increase in CRR and SLR will decrease money multiplier. With the same monetary base, for implementation of a restrictive monetary policy the RBI will increase the reserve deposit ratio.

Money multiplier is generally stable under normal periods, but not always. For instance, during the early part of the Great Depression (1929 to 1933), there was significant decline in the value of the money multiplier in the US. Hence, depositors became nervous about the financial condition of banks, and withdrew their deposits. It led to an increase in cd ; simultaneously it compelled banks to hold more reserves which increased rd .

15.3.2 An Alternative Interpretation of Money Multiplier

You may recall that money multiplier is defined as the ratio $\frac{M1}{M0}$. From equation (15.1) we know that $M1 = (cd + 1)D$. Now let us denote the public's desired currency-money ratio as $cm = \frac{C}{M1}$. By substituting the value $M1 = (cd + 1)D$ from equation (15.1), we obtain

$$cm = C/(cd + 1)D = (C/D)/(cd + 1)$$

Thus we have

$$cm = \frac{cd}{cd + 1} \quad \dots(15.4)$$

$$\text{and, } (1 - cm) = 1 - \frac{cd}{cd + 1} = \frac{1}{cd + 1} \quad \dots(15.5)$$

Substituting the above values in (15.3), we obtain

$$M1 = \frac{cd + 1}{cd + rd} M0 = \frac{1}{\frac{cd}{cd + 1} + \frac{rd}{cd + 1}} M0 = \frac{1}{cm + rd(1 - cm)} M0 = \frac{1}{1 - (1 - rd)(1 - cm)} M0 \quad \dots(15.6)$$

The monetary base is controlled by the central bank through variations in CRR and SLR. Apart from the above, there are channels such as open market operations, interest rates, and exchange rates.

To understand the mechanism of the effect of open market operations on the money supply, let us assume that the RBI increases M_0 by the amount ΔM_0 through open-market operations such as buying bonds. A seller of the bonds keeps the fraction cm as cash and deposits the remaining portion, $(1 - cm)$, at the bank. At the second round, the bank keeps the fraction rd of $(1 - cm)\Delta M_0$ as reserves. The remaining amount, i.e., $(1 - rd)(1 - cm)\Delta M_0$ is given as loans to customers or buys bonds from the RBI. The money supply will be further increased by $(1 - rd)^2(1 - cm)^2\Delta M_0$ in the third round, and so on.

If cm and rd are constant (for simplicity), the total increase in money supply at the end would be:

$$\Delta M_1 = \Delta M_0 + (1 - rd)(1 - cm)\Delta M_0 + (1 - rd)^2(1 - cm)^2\Delta M_0 + \dots = \frac{1}{1 - (1 - rd)(1 - cm)}\Delta M_0 \quad \dots (15.7)$$

(Equation (15.7) can be seen as an infinite geometric series. We apply the algebraic rule of sum of geometric series.)

We can conclude that an increase in monetary base by a specific amount leads to an increase in the money supply by m times of the increase in the monetary base.

Check Your Progress 2

- 1) Define money multiplier. Derive the money multiplier in terms of desired currency-deposit ratio and desired reserve-deposit ratio.

.....

.....

.....

.....

- 2) What is the non-bank public's desired currency-money ratio? Explain how it is related to the money multiplier.

.....

.....

.....

.....

15.3 OPEN MARKET OPERATIONS

As we have discussed earlier, there are four main functions of a central bank.

1. They act as the banker's bank or 'lender of last resort'. They are responsible for maintaining solvency and liquidity of the commercial banks.

2. They promote stable economic growth.
3. They act as the stabiliser of the purchasing power of the currency, or maintain stable inflation.
4. They manage the exchange rate in the market to reduce the volatility.

While these functions are complementary, there could be situations when one function conflicts with another. For instance, controlling of inflation can be sometimes achieved at the cost of economic growth. Thus when the objective is 'inflation targeting', economic growth may be somewhat lower. Similarly, in times of crisis the central bank raises the money supply to maintain liquidity in the economy and it creates inflationary pressure. The central banks work to achieve certain goals mainly targeting the interest rates, money supply targets and exchange rates, which are done through the monetary policy such as Open Market Operations (OMO), changes in reserve requirements, or changes in the bank rate.

Let us focus on the OMO. This is a popular policy tool available to central banks to influence the monetary base. Through this policy, the central bank buys government securities from the commercial banks and other financial institutions with an objective of increasing the monetary base. Such a policy stance is called expansionary monetary policy. The central bank makes payment for these securities in the form of reserves credited to the account held by the commercial banks with the central bank. It leads to a change in the composition of assets of the balance sheet of the commercial bank. It lowers the income producing assets like securities and increases the non-income producing assets like excess reserves of the commercial banks. Then, the commercial banks would try to convert a part or all of these excess reserves to provide new loans by lowering its interest rate. The central banks adopt expansionary monetary policy in a situation of higher demand for liquidity to keep the interest rates in check.

In contrast, the central bank opts for a contractionary monetary policy if its objective is to reduce the monetary base. In this case, the central bank sells the government securities to the commercial banks or other institutions to siphon-out some money from circulation. This process of increasing or reducing the monetary base by using the expansionary (contractionary) monetary policy, through buying (selling) of the government securities is known as the Open Market Operations. Recall that change in monetary base leads to change in money supply through the money multiplier.